

# Building a unified science of sensorimotor control with Densely Observable Measurement Environments [DOMEs]

FreeMoCap Foundation, Inc.

Written Proposal to the NSF X-Labs Initiative

## 1. Mission

The MISSION of the **FreeMoCap Foundation X-Lab (FMC-X)** is to develop and disseminate the tools to measure a new scientific observable representing the closed loop of the **perceptual, mechanical, and environmental** aspects of goal-directed behavior of freely moving organisms to reshape perceptual/motor neuroscience, musculoskeletal biomechanics, embodied/agentive artificial intelligence, and mobile robotics into a new convergent science of **sensorimotor control in natural environments**.

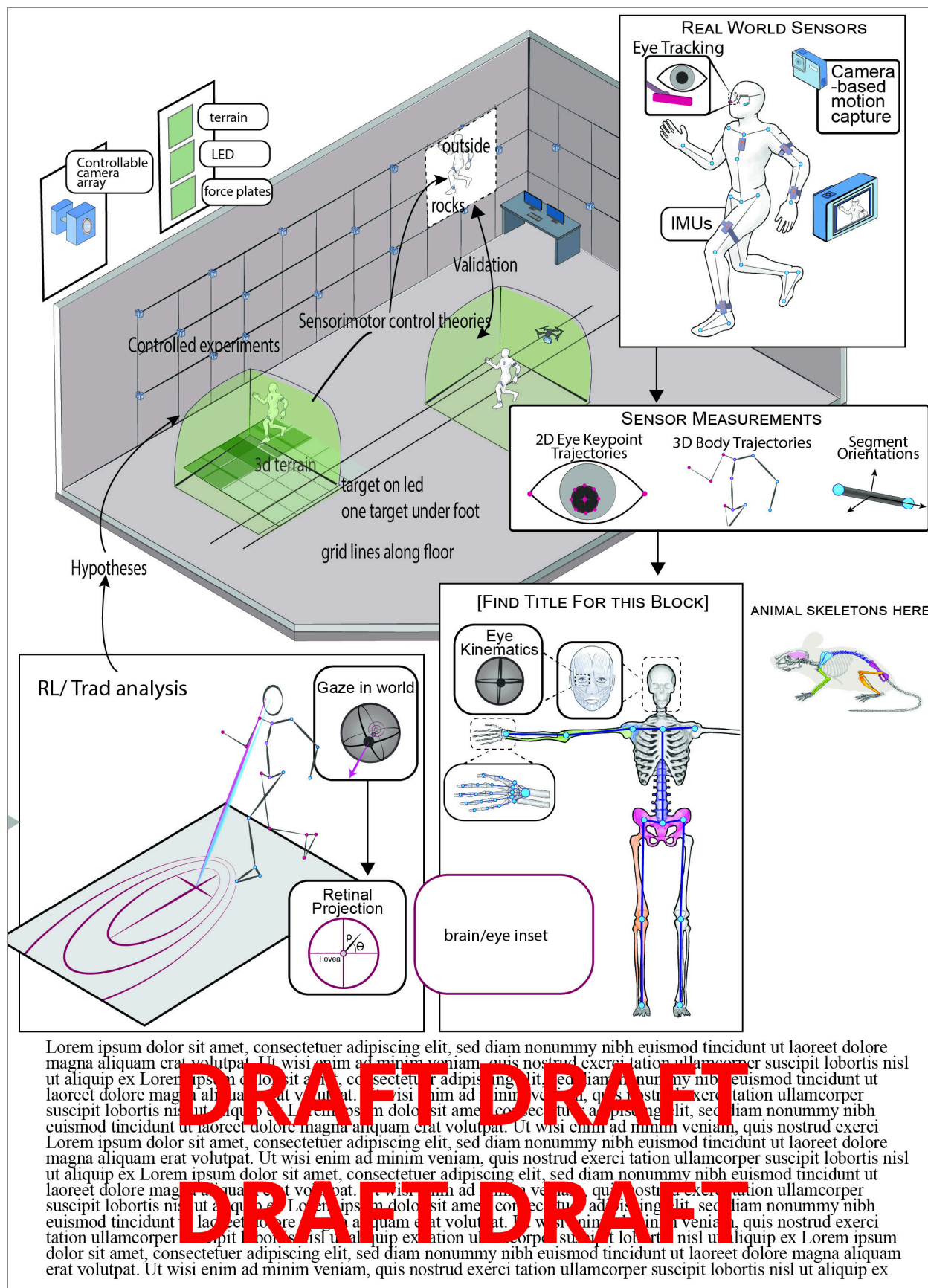
### 1.1. Measurement target

We will organize our technical development around the technical bottlenecks which limit our ability to combine reconstructions the **eye, body, and environment** into combined representations which trace to activity in the central and peripheral nervous system ...

### 1.2. Flagship

We will develop and validate a **novel research instrument** we call a **DOME** (*Densely Overlapping Measurement Environment, Ref - Figure 1 - Hero Figure showing full DOME*), which unifies a heterogeneous array of multi-modal sensors into single, centrally-controllable, metrologically-validated instrument for the measurement and manipulation of a freely moving agent within a specified 3D volume of real-world space.

- **DOME-L** — the flagship instrumented volume, in the greater Boston area. Large enough to fully enclose the smaller variants for validation and metrological grounding.
- **DOME-S** — the extension of the commodity-camera capture volumes already built and disseminated through the FreeMoCap Project. Representative of the DOME a research lab or a classroom would build.
- **DOME-Mobile** — a wearable, self-contained DOME producing Dense Observations in unconstrained indoor and outdoor environments, continuing the lineage of the PI's gaze/gait and retinal optic flow work<sup>1-4</sup>.



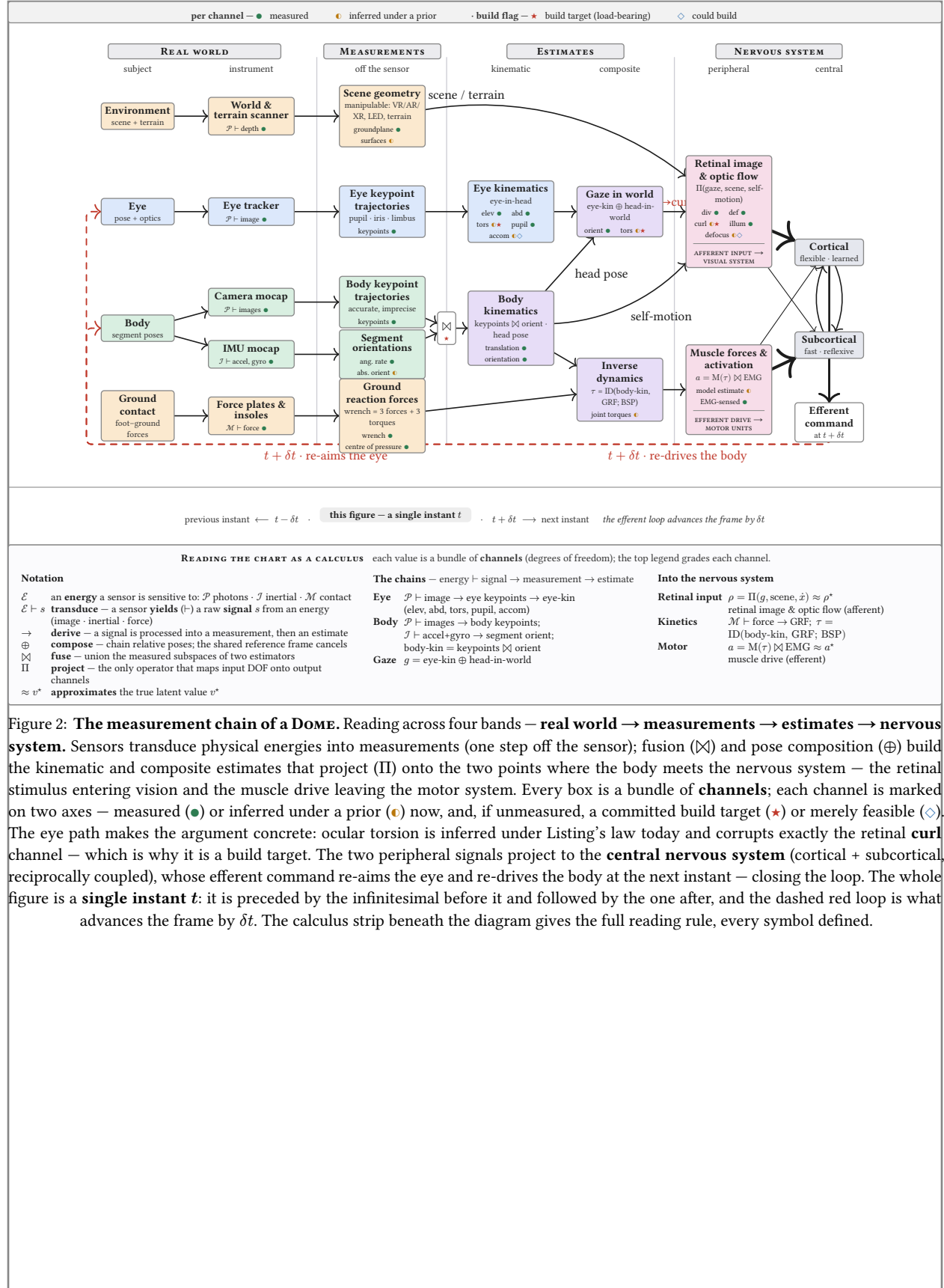


Figure 2: **The measurement chain of a DOME.** Reading across four bands — **real world** → **measurements** → **estimates** → **nervous system**. Sensors transduce physical energies into measurements (one step off the sensor); fusion ( $\boxtimes$ ) and pose composition ( $\oplus$ ) build the kinematic and composite estimates that project ( $\Pi$ ) onto the two points where the body meets the nervous system — the retinal stimulus entering vision and the muscle drive leaving the motor system. Every box is a bundle of **channels**; each channel is marked on two axes — measured (●) or inferred under a prior (●) now, and, if unmeasured, a committed build target (★) or merely feasible (◇). The eye path makes the argument concrete: ocular torsion is inferred under Listing's law today and corrupts exactly the retinal **curl** channel — which is why it is a build target. The two peripheral signals project to the **central nervous system** (cortical + subcortical, reciprocally coupled), whose efferent command re-aims the eye and re-drives the body at the next instant — closing the loop. The whole figure is a **single instant**  $t$ : it is preceded by the infinitesimal before it and followed by the one after, and the dashed red loop is what advances the frame by  $\delta t$ . The calculus strip beneath the diagram gives the full reading rule, every symbol defined.

**2. Technology Landscape**

- Existing tools and methods
  - Myriad single-modality hyperspecialized research threads
    - OpenSim - Widely used but universally hated by practitioners
    - DeepLabCut - Global success on paper, universally a headache in practice
    - OpenCap works, but only for iPhones and requires a cloud upload, and caters exclusively to clinical biomechanics
    - SIMPL/Meshcapade - Has built past empirically validity (3d from 2d) pipelines not truth preserving. Building increasingly more abstract and less traceable data channels on top of the same 2d data we've had for a while
    - etc
  - MuJoCo/IsaacLab Driving robotics
    - Integrate NeuroBehavioral Data into those simulation/RL training environments to cross-align perceptuomotor neuroscience, HNHA biomechanics, robotics control research (e.g. Cassie from Guinea Fowl)
- New approaches to complexity management
  - Ontology/Entity Management (Anduril/Palantir/Maven Smart System)
  - Industry and Military technology applied to scientific domain
  - Want to establish an operational ontology as backbone of this tool, following powerful patterns established by (Anduril/Palantir/Maven) systems
- FreeMoCap and its landscape
  - Established Tool with Growing International Community
  - Existence is proof-of-concept of approach, XLabs scale support would accelerate FreeMoCap model into the global scale
- Laser Skeleton research
  - <sup>1-3</sup>
  - etc
  - Ferret work
  -

**3. Outcomes**

- Cool Tool - ~~Skinner~~SkellyBox: Multiscale modular empirical capture arena
  - Interoperability across species, domains, and timescales
    - A member of a mouse lab can walk into a marmoset lab and expect their pipelines will still work
    - Perceptuomotor behavior and kinematic/kinetic analysis directly interoperable with robotics models (via MuJoCo and Nvidia Isaac Gym) - Data for reinforcement learning models, Controls to interpret HNHA behavioral data
    - Data models alignment across timescales - developmental/longitudinal (months/years/lifespan) timescales aligned with perception/action timescales (seconds/minutes)
  - Composable complexity
    - Careful ontology allows for complexity via composable data sources and associated downstream entities
    - Adding new sensor modalities into established scene, framework handles spatiotemporal alignment/calibration, unit conversions, resampling, etc
- Organize and operationalize a new culture of science around discussion/management/extension shared platform tool
  - In the form of an annual Congress meeting
  - Organizing the culture of science around the discussion and advancement of shared ontologies rather than an endless stream of 8-10 page pdfs
  - Break the Skill Ceiling in academic science
    - Build tool to allow multi-decade continuous education across disciplines and experience (same tool in high school as used in federally funded research lab)
    - Shared toolset encourages natural blending of disciplines while encouraging and reward higher specialized exploration (esp when that research produces actionable results across different dimensional domains)
- Mesoscale revolution
  - Allow advances in reductionist microscale research to align with human-scale interests
  - Align Perceptuomotor Neuroscience with humanoid robotics, allow transfer, technology, and technique transfer between those fields

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| 3.1. Timeline |           |                                                                                                                                                                                                                                                                                                                                     |
|---------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase         | Name      | Description                                                                                                                                                                                                                                                                                                                         |
| 1             | Organize  | Set up Organization (Phase 0) – Solidify release of FreeMoCap v2                                                                                                                                                                                                                                                                    |
| 2             | Architect | Develop core software architecture, define key ontologies and entity/action pairings via use-oriented development (ferret, mouse, guinea fowl, marmoset, human research). Design and prototype instrumentation (mocap/eye cameras, electrode arrays and miniscopes, motor-unit EMG, environmental control systems)                  |
| 3             | Execute   | Integrate new architecture into global/public facing community of users (presented as FreeMoCap v3). Iterative design of instrumented system                                                                                                                                                                                        |
| 4             | Stabilize | Establish OODA loop within expected standard operations                                                                                                                                                                                                                                                                             |
| 5             | Expand    | Once stabilized, expand key performance metrics                                                                                                                                                                                                                                                                                     |
| 6             | Align     | Focus on building deeply integrated community of users and contributors (Transition BDFL → BDFaSPoT)                                                                                                                                                                                                                                |
| 7             | Establish | Focus on long term stability and healthy growth                                                                                                                                                                                                                                                                                     |
| 8             | Maintain  | Beyond proposal timeline – Establish organization practice in service of unbounded infinite timeline. Apply winning model to support extending ‘mesoscale research revolution’ to all biological sciences (e.g. integrating deeper microbio/biochem insights beyond the scope of the present perceptuomotor/biomech/robotics focus) |

Table 1: Timeline of Phased Milestones

4. Senior/Key Personnel Qualifications

4.1. Jonathan Matthis, PhD [Principle Investigator]

4.2. EI

4.3. AC, PhD

4.4. RR, CPA or w/e

| Name / Role                        | Qualifications                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jonathan Matthis, PhD<br>President | <ul style="list-style-type: none"><li>• Founder and chief maintainer of FreeMoCap project</li><li>• Left tenure track academic research position precisely because the institution could not support the work that needed to be done</li><li>• Decades experience integrating visual neuroscience with biomechanics/robotics<sup>1-3</sup></li></ul>                                                                                      |
| NR,<br>CEO                         | <ul style="list-style-type: none"><li>• Expertise in technically and software dev</li><li>• Previous CEO XP, founded and ran own buisness</li><li>• CEO helps PI apply his specific expertise to the core tasks of the XLab</li></ul>                                                                                                                                                                                                     |
| EI<br>CTO                          | <ul style="list-style-type: none"><li>• Decades of experience in software Industry</li><li>• Chief architect and CTO for established companies</li><li>• Represents expertise truly absent from academic contexts</li><li>• Ensures world-class enterprise-level software capability</li><li>• Will build training/workshop/hackathon backend to inject EI-level software development expertise into broad community of science</li></ul> |
| AC, PhD<br>CSO                     | <ul style="list-style-type: none"><li>• Expertise in validation of clinical research tools and bench-to-bedside development</li><li>• [CITE DISSERTATION]</li><li>• Experience in neuroprosthetics design and brain imaging</li><li>• Worked with PI on core FMC dev and validation</li></ul>                                                                                                                                             |
| RR<br>CFO                          | <ul style="list-style-type: none"><li>• Bookkeeping</li><li>• NSF Grant management</li><li>• Finances</li><li>• Entrepreneurship (founded own Bookkeeping coop)</li><li>• Taught Finance Theory at university level</li></ul>                                                                                                                                                                                                             |
| KM<br>Project Manager (?)          | <ul style="list-style-type: none"><li>• Key driver of<sup>3</sup></li><li>• Worked with PI on core tech<sup>2</sup></li><li>• Expertise in key technologies</li></ul>                                                                                                                                                                                                                                                                     |
| MN<br>Project Manager (?)          | <ul style="list-style-type: none"><li>• Expertise in clinical biomechanics,</li><li>• Medical systems design,</li><li>• Mechatronics for clinical tools,</li><li>• Multi-modal motion capture lab management</li></ul>                                                                                                                                                                                                                    |

Table 2: Senior/Key Personnel Qualifications

**5. Team Capabilities Statement****5.1. Senior Team Capabilities**

- Jon - done it before in humans, doing it in ferrets and mice
- AC - focusing clinical adoption and bench-to-bedside transitions
- EI - Deep professional experience in Tech Sector, high level software architecture experienced necessary to develop truly global scale enterprise level of software in open-source/academic contexts
- RR - Organizational health and entrepreneurial development

**5.2. Collaborators**

| Initials           | Model / Domain       | Expertise                                  |
|--------------------|----------------------|--------------------------------------------|
| BS                 | Ferrets              | Visual/Perceptual neuroscience, ephys      |
| MD                 | Guinea Fowl          | Musculoskeletal Biomechanics, muscle units |
| DF                 | Mice                 | System Biology                             |
| AH                 | Marmosets / NHP      | Ephys                                      |
| MH                 | Human                | Vision and CPS                             |
| Mayo Clinic        | Human                | Brain scans and implanted sensors          |
| BD, GN/RAI, CH, JH | Robots / Fabrication | Control theory                             |
| AA                 | Physical facilities  | Rapid iteration, recruiting                |

Table 3: Collaborator Network

**5.3. Governance**

- Most successful FOSS project operate under a Benevolent Dictator For Life (BDFL) Governance (e.g. Linus Torvald at Linux)
- This is best when an organization is young, but later on centralization is a concern control should melt into the community of users (see Ton Roosendaal's step down from Blender lead position)
- We will use Benevolent Dictator for a Set Period of Time (BDfaSPoT)
  - Transition from centralized control: President -> President + Board -> to President + Board (C-Suite) + Community DAO
    - President maintains at least 51% control until year 5, then optional transition to 50/50 President/Board+Community
- Plan changes via public Requests for Comments (similar to Python PEP system)
  - Provide public space for feedback on planned features, allows transparency without creating a bottleneck
- In 2nd year, create Developer Fund and Community Grants program
  - Stakeholders vote on proposals and features, via DAO like voting
- In 4-5th year, extend this system to handle community steering of org

**5.4. Autonomy**

- We are already autonomous, as of 2 weeks prior to the deadline of this proposal.
  - This project will happen no matter what, but with XLabs funding, we could truly change the world

**Bibliography**

1. Matthis, J. S., Yates, J. L. & Hayhoe, M. M. Gaze and the Control of Foot Placement When Walking in Natural Terrain. *Current Biology* **28**, 1224–1233.e5 (2018).
2. Matthis, J. S., Muller, K. S., Bonnen, K. L. & Hayhoe, M. M. Retinal Optic Flow during Natural Locomotion. *PLOS Computational Biology* **18**, e1009575 (2022).
3. Muller, K. S. *et al.* Retinal Motion Statistics during Natural Locomotion. *eLife* **12**, e82410 (2023).
4. Muller, K. S. *et al.* Foothold Selection during Locomotion in Uneven Terrain: Results from the Integration of Eye Tracking, Motion Capture, and Photogrammetry. <https://elifesciences.org/reviewed-preprints/91243v2> (2024) doi:10.7554/eLife.91243.2.